CASE STUDY

# SupportOps AI

A multi-agent customer support platform where AI agents triage and draft responses, but a deterministic policy engine — not the model — decides what's safe to send automatically.

> **Self-directed portfolio project.** Not a production deployment for a real company. It was built to practice a genuinely hard distributed-systems problem — orchestrating stateful AI agents safely — rather than a toy chatbot demo. Every technical claim below is verified directly against the current codebase and its 167-test suite, and every open gap is stated plainly rather than glossed over — see Resources.

FastAPI | LangGraph | CrewAI | PostgreSQL | Redis | 167 Tests Passing

1 · THE PROBLEM

## Support triage is repetitive, but the risky 5% can't be automated blindly

Support teams field a high volume of billing, technical, and account questions that follow predictable patterns — good candidates for AI triage and drafting. But a fraction of tickets involve refunds, legal threats, security incidents, or fraud, where an autonomous AI response is the wrong failure mode entirely. The problem isn't "can an LLM answer support tickets" — it's building a system that knows the difference and routes accordingly, every time, with a record of why.

- Classifying and routing tickets to the right specialist consistently
- Keeping specialist answers grounded in real product knowledge, not invented
- Recognizing high-risk tickets (refund, legal, security, fraud) and holding them for a human
- Making retries and duplicate submissions safe instead of silently creating duplicate tickets
- Producing an audit trail for every automated and human decision

2 · OBJECTIVES

## Real reasoning, deterministic guardrails

The system had to keep two things separate on purpose: what the AI is allowed to decide, and what a policy engine decides for it.

- **Real specialist reasoning** — CrewAI agents grounded in an actual knowledge base, not scripted replies
- **Deterministic escalation** — a policy engine, not the model, decides what needs human approval
- **Idempotent intake** — a retried request must never create a duplicate ticket
- **Postgres as the system of record** — Redis can fail without losing business data
- **A full audit trail** — every automated action and every human approve/edit/reject is logged

| 4 | 6 | 8 | 167 |
|---|---|---|---|
| specialist agents: billing, technical, account, general | node LangGraph workflow, fixed topology | policy triggers routing tickets to human review | automated tests across unit + integration |

3 · SOLUTION

## LangGraph owns control flow; CrewAI owns reasoning

LangGraph compiles a fixed six-node graph that owns ticket state, classification, routing, and escalation. CrewAI supplies the domain specialists it calls into — each grounded by a Postgres-backed knowledge tool — but a specialist never controls the workflow or decides its own escalation; it can only report that it's unresolved, and the policy engine decides what happens next.


*The graph's topology has stayed fixed since it was first built — every node went from placeholder to a real implementation without changing the shape of the workflow.*

4 · TECHNOLOGIES USED

## Stack

- **FastAPI** — HTTP layer, JWT/OAuth2-protected routes
- **LangGraph** — Stateful workflow orchestration
- **CrewAI** — Role-based specialist agents
- **OpenAI** — Structured-output classification + agent reasoning
- **PostgreSQL** — System of record — tickets, approvals, audit logs
- **Redis** — Rate limiting, idempotency, caching, checkpoints
- **SQLAlchemy + Alembic** — Async ORM + real per-table migrations
- **pytest** — 167-test suite, unit + integration

5 · CHALLENGES SOLVED

## The distributed-systems edges, not the chatbot part

### Two event loops, one shared client

FastAPI's request loop and a dedicated background loop bridging LangGraph's synchronous nodes each need their own Redis/Postgres client — sharing one across both used to crash with a `RuntimeError`, and for Postgres specifically could poison a pooled connection so a later, unrelated request would 500. Fixed by handing out one client/engine per calling event loop.

### Choosing failure mode per feature, not uniformly

Caching and workflow checkpoints fail *open* — a miss just recomputes. Idempotency fails *closed* — `POST /tickets` returns 503 if Redis is unreachable, because silently proceeding could let a duplicate ticket through. Same infrastructure, deliberately different guarantees per feature.

### Keeping agents grounded, not freelancing

Specialist agents answer only from a Postgres-backed knowledge-base tool (Redis-cached), not from open-ended model knowledge — so an answer can be traced back to a real reference article instead of the model's own guess.

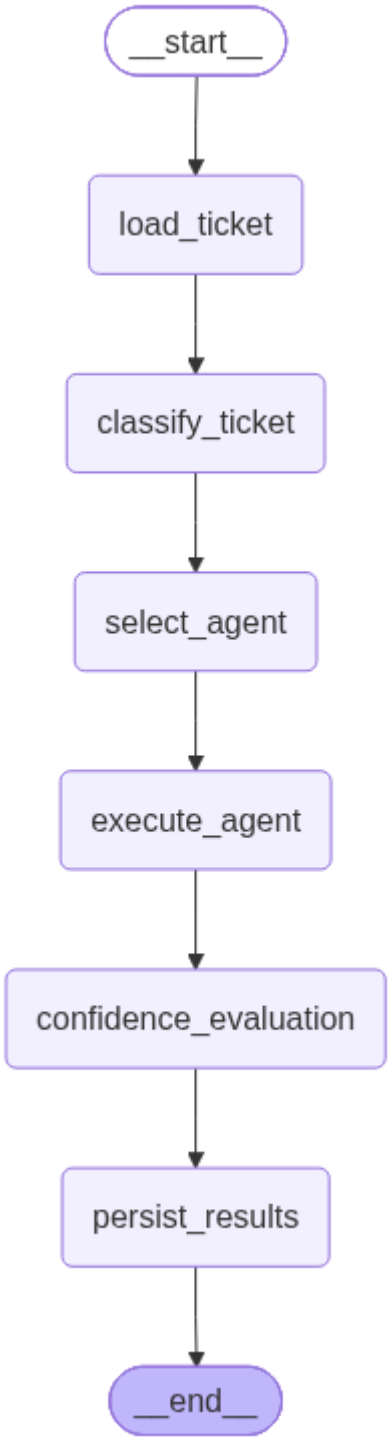
### Separating "AI reports" from "policy decides"

A CrewAI agent can report that it's unresolved, but it never decides escalation itself. That decision belongs entirely to `policy/rules.py`'s deterministic keyword/threshold checks — an explicit design boundary between reasoning and control.

6 · ARCHITECTURE

## Postgres is the system of record; Redis is disposable

PostgreSQL holds every piece of business data — tickets, approval requests, audit logs, users. Redis only ever holds data that can safely expire or be rebuilt: rate-limit counters, idempotency claims, workflow checkpoints, and AI/knowledge-base caches. A Redis outage never puts business data at risk — the one exception is idempotency, which fails closed by design (see Section 5).




*The compiled LangGraph workflow graph, rendered via* `backend.scripts.export_workflow`.

7 · AI WORKFLOW

## Classification, reasoning, and escalation are three separate steps

**classify_ticket**
OpenAI structured output
→
**execute_agent**
CrewAI + KB tool
→
**confidence_evaluation**
policy/rules.py
→
**Supervisor Queue**
if flagged

***Guardrail:*** *escalation triggers on deterministic keyword/threshold rules (refund, legal, lawsuit, attorney, security, breach, fraud, low-confidence) plus a structured* `agent_unresolved` *signal the agent reports — never a decision the model makes for itself. Flagged tickets become a real* `ApprovalRequest` *row a Supervisor/Admin can view, edit, approve, or reject, each action writing its own audit log row.*

8 · TESTING

## 167 tests, real integration coverage included

159 unit tests run fully mocked (Redis via `fakeredis`, Postgres via test doubles) and finish fast with no external services. 8 integration tests exercise real Redis and/or Postgres — including the cross-event-loop regression from Section 5 — and skip themselves automatically if the service they need isn't reachable, so a plain `pytest` run is always safe.

| Area | What's verified |
|---|---|
| Graph nodes | classify, execute_agent, persist_results, checkpointing behavior |
| Policy engine | Escalation rules and adversarial-input handling |
| Supervisor API | Queue listing, approve/edit/reject, adversarial cases |
| Services | Idempotency store, audit logging, ticket creation + execution |
| Core infra | Rate limiting, caching, async utils, Redis client lifecycle |
| Integration | Real Redis+Postgres ticket workflow, cross-loop client isolation, live schema-vs-model drift |

9 · RESULTS

## What's verified vs. what's still open

**167 / 167**
automated tests passing
VERIFIED

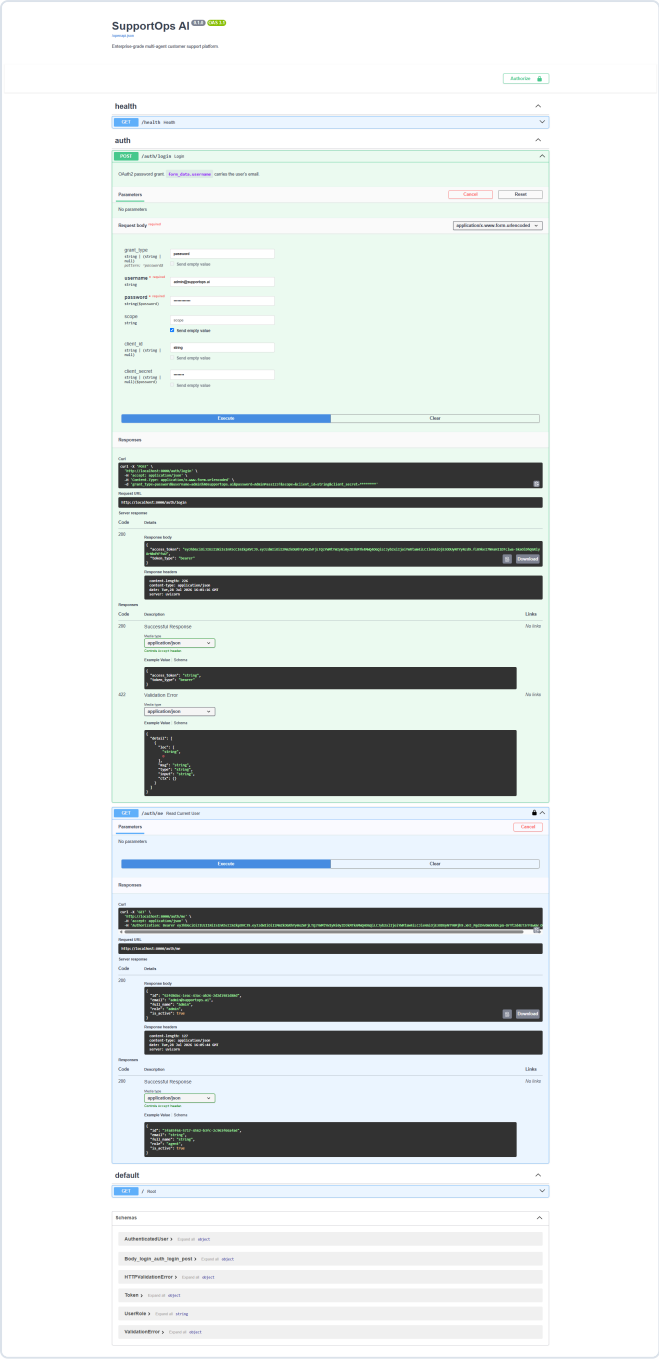
**5**
endpoints backed by real Postgres logic, not stubs
VERIFIED

**1**
confidence score still a fixed placeholder, not model-derived
OPEN GAP

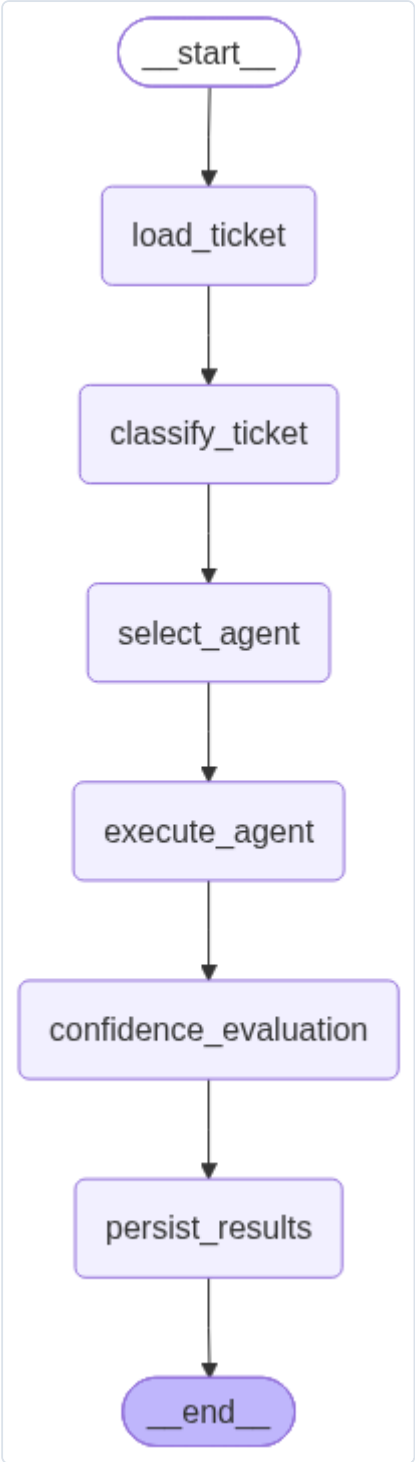
**0**
tenants — single-tenant policy thresholds today
OPEN GAP

*This is a portfolio build, not a live deployment — there's no production ticket volume to report. The open gaps above are called out on purpose; see Section 2 of the Technical Architecture doc for the full status table.*

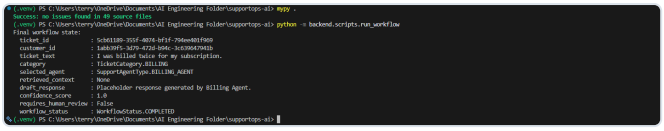
# Evidence, not mockups



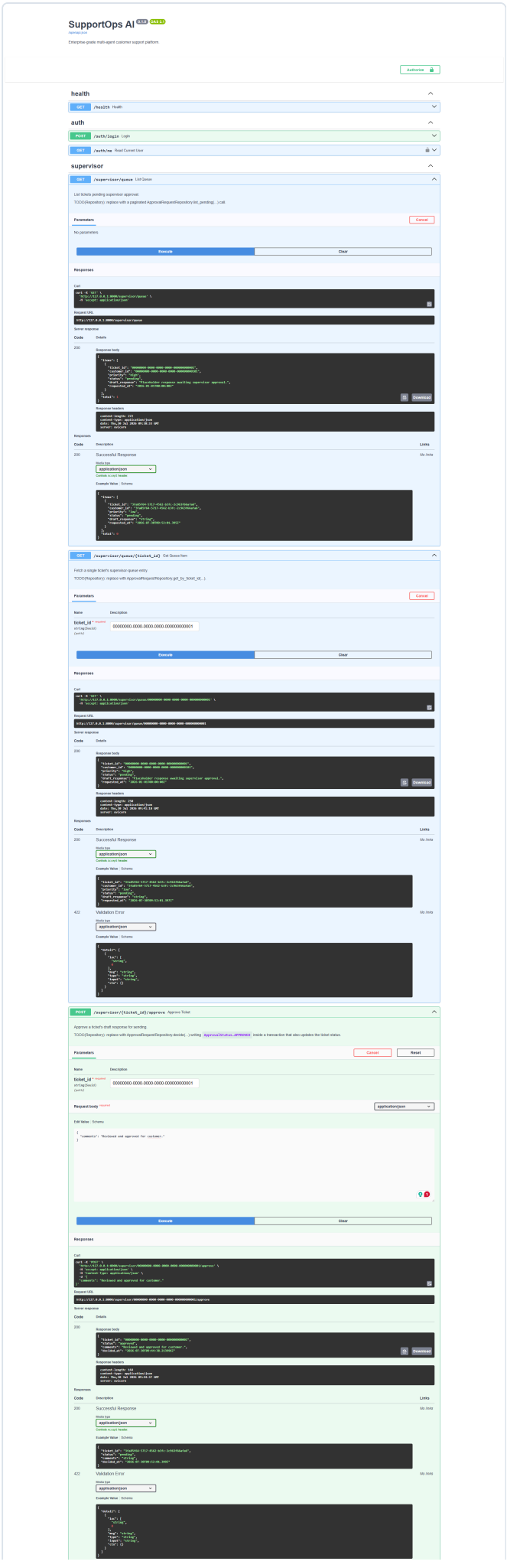

**Authentication flow** — JWT login via Swagger UI

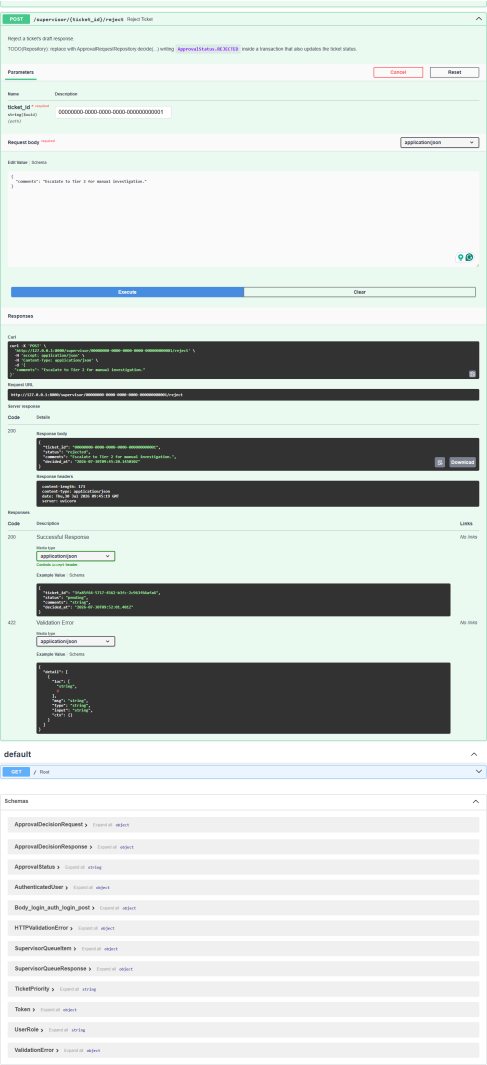



**Workflow graph** — the compiled LangGraph pipeline




**LangGraph smoke test** — a real ticket run through the graph





**Supervisor API** — approval queue smoke test

11 · KEY TAKEAWAYS

## What this build was really practicing

1. Separating "the AI decides" from "policy decides" at a hard boundary is what makes autonomous agent behavior safe to ship.
2. Ephemeral infrastructure (Redis) and system-of-record data (Postgres) need explicitly different failure modes, chosen per feature.
3. Grounding agents in a real retrieval tool, not open-ended model knowledge, is what makes their answers auditable.
4. Multi-event-loop bugs are invisible until you specifically test the interaction — a smoke test alone wouldn't have caught the cross-loop connection poisoning.
5. An idempotent write path is a correctness requirement, not a nice-to-have, the moment retries are possible.
6. Naming a fixed-value placeholder honestly ( `_PLACEHOLDER_CONFIDENCE_SCORE` ) in the code and the docs beats quietly presenting it as calibrated.

## Resources

Want to verify or explore this further? Here you go.

**GitHub repository**
Full source, commit history, 167 passing tests
github.com/Terrytd0/Supportops-AI

**README**
Setup, feature list, architecture overview
README.md

**Demo video**
Live walkthrough of the workflow end to end
Watch on Google Drive

**ADR-001**
LangGraph vs. CrewAI, and why the project uses both
docs/adr/ADR-001-langgraph-vs-crewai.md

**Technical Architecture**
The companion document to this case study
technical-architecture.pdf

**Full docs folder**
Architecture, requirements, ADRs, design review
docs/